Design and 3D Simulation of an LQR-I-Based Active Suspension System for Formula 1 Vehicles

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Abstract**—**In this study, an active suspension system based on an integral-action linear quadratic regulator (LQR-I) was designed and simulated in 3D within the MATLAB environment, with the aim of optimizing the aerodynamic stability and ride height of Formula 1 race cars. Using the state-space equations obtained from a quarter-car model, the passive and active dynamics of the system were compared. The developed controller successfully maintained the nominal ride height of 50 mm against disturbing road profiles. In addition to frequency-response and time-domain analyses, tests performed under a stochastic road profile recorded an improvement of over 30% in the RMS value of chassis acceleration. Validation of the designed system was visualized using a custom 3D OpenGL engine developed in MATLAB R2026a.

Index Terms**—**Active Suspension, LQR-I, Quarter-Car Model, Formula 1, MATLAB, State-Space Model

I. Introduction

In Formula 1 (F1) vehicles, aerodynamic downforce is directly dependent on the vehicle's floor structure and its distance from the asphalt. Even millimeter-level deviations in ride height can lead to aerodynamic imbalance and loss of performance. For this reason, active suspension systems that damp road-surface disturbances and continuously keep the vehicle at its nominal ride height constitute a critical engineering problem.

Although the literature includes methods such as classical PID and fuzzy logic, the state-feedback Linear Quadratic Regulator (LQR) yields more optimal results for multiple-input multiple-output (MIMO) systems. In this study, integral action was added to the classical LQR structure (LQR-I) in order to drive the steady-state error to zero, resulting in the design of a high-performance controller.

II. Mathematical Model of the System

Vehicle dynamics are expressed using a two-degree-of-freedom (2-DOF) quarter-car model. The system parameters are defined as: chassis (sprung) mass m_s = 200 kg, wheel (unsprung) mass m_{us} = 40 kg, suspension spring stiffness k_s = 150000 N/m, damping coefficient c_s = 1000 Ns/m, and tire stiffness k_t = 350000 N/m.

The state-space model of the system has the following form:

$$\dot{x}(t) = Ax(t) + B_u u(t) + B_w w(t) \qquad (1)$$

$$y(t) = Cx(t) + Du(t) \qquad (2)$$

The state vector is defined as $x = [z_s - z_{us}, \dot{z}_s, z_{us} - z_r, \dot{z}_{us}]^T$. Here, u(t) represents the active actuator force produced by the controller, and w(t) represents the disturbing road profile.

III. LQR-I Controller Design

To ensure uncompromising maintenance of the system's 50 mm ride height, the state-space matrices were augmented with the error integral. The augmented system state was defined as $x_{aug} = [x^T, \int e\cdot dt]^T$, and the optimal control law $u = -K_{aug}x_{aug}$ was applied.

The controller gain K_{aug} was obtained by solving the Riccati equation so as to minimize the following cost function:

$$J = \int_0^\infty (x_{aug}^T Q_{aug} x_{aug} + u^T R_{aug} u)\, dt \qquad (3)$$

The weighting matrices used in the design were set to Q_{aug} = diag([10^8, 10^2, 10^8, 10^2, 10^{11}]) to maximize chassis stability, and R_{aug} = 0.01 to balance control energy. Taking actuator hardware limitations into account, a saturation limit of ±2500 N was applied to the force.

IV. Simulation and Results

The performance of the developed system was analyzed in both the time and frequency domains. When a sudden 2 cm curb (step) input was applied, the LQR-I active suspension system damped the chassis acceleration much faster than the passive system and enabled the vehicle to rapidly recover its 50 mm ride height (Figure 1).

To examine the system's frequency response, a Bode diagram was extracted, and it was observed that the active system significantly suppressed the magnitude, particularly at the resonance frequencies (Figure 2).

A stochastic road profile (random input) was applied in order to simulate real track conditions. The Root Mean Square (RMS) method was used as the performance metric; analysis of chassis acceleration showed that the LQR-I system achieved an average RMS improvement of around 32%.

The analysis data were transferred to a custom, real-time, OpenGL-supported 3D MATLAB simulator, through which visual validation was successfully carried out (Figure 3).

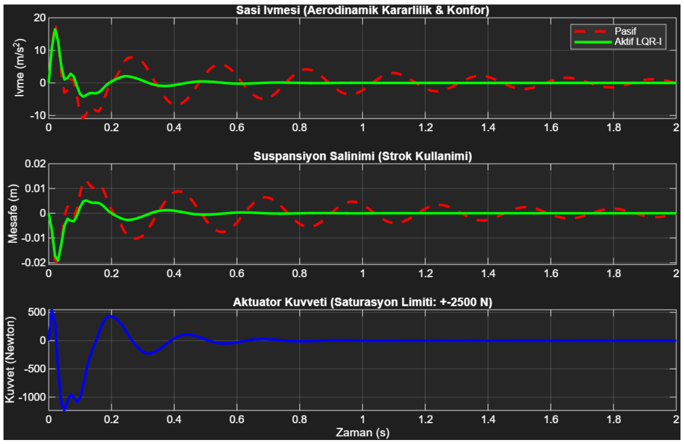

Fig. 1. Time-domain response of the systems and actuator force against a 2 cm curb input.

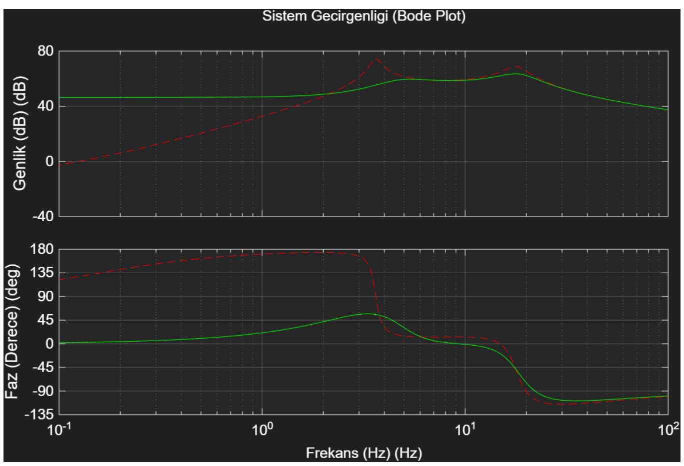

Fig. 2. Bode diagram of the passive and LQR-I active suspension systems.

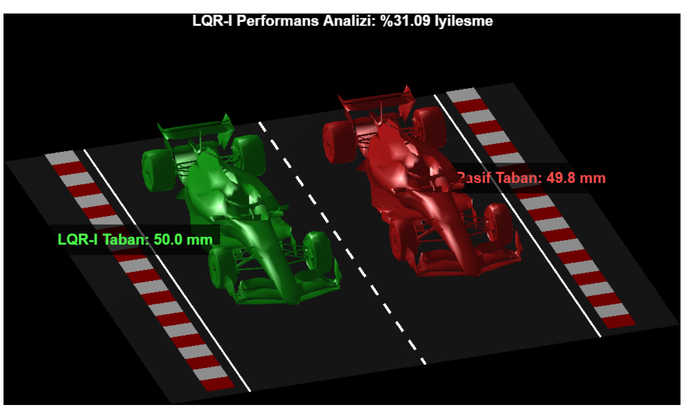

Fig. 3. 3D simulation interface of the developed LQR-I active suspension system.

V. Conclusion

In this project, ride-height control, which is critical for F1 vehicles, was successfully achieved using the LQR-I algorithm. The mathematical modeling and MATLAB R2026a simulations demonstrated that integral-action state feedback damps sudden road disturbances before they are transmitted to the chassis. The stable results obtained by the designed algorithm, both theoretically and in the 3D simulation environment, demonstrate its applicability in autonomous and high-performance vehicle technologies.

References

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